

Order Processing System – Workflow & Rules

This document describes a simplified order processing system used in an e-commerce platform. It includes entities, relationships, process steps, decision points, and business rules.

Entities:

- 1 Customer: A user placing an order
- 2 Order: A collection of items requested by the customer
- 3 Item: A product available for purchase
- 4 Payment: Transaction made by the customer
- 5 Inventory: Stock levels for items
- 6 Shipment: Delivery process for orders

Relationships:

- 1 A Customer places an Order
- 2 An Order contains multiple Items
- 3 An Order requires a Payment
- 4 Inventory tracks availability of Items
- 5 Shipment fulfills an Order

Process Steps:

- 1 1. Customer places an order
- 2 2. System validates order details
- 3 3. Check inventory availability
- 4 4. Process payment
- 5 5. Confirm order
- 6 6. Initiate shipment

Decision Points:

- 1 If payment fails → cancel order
- 2 If item is out of stock → notify customer
- 3 If address is invalid → request correction

Business Rules:

- 1 Orders above \$500 require manual review
- 2 Payments must be completed within 15 minutes
- 3 Inventory must be updated immediately after order confirmation
- 4 Shipments must be initiated within 24 hours

Example Scenario:

A customer places an order for 2 items. The system validates the order and checks inventory. If both items are in stock, payment is processed. If payment succeeds, the order is confirmed and shipment is initiated. If any step fails, the system triggers the corresponding decision rule.